Step 3 stats:
Variant: 3653_C_T (C→T at position 3653)
Gene: ORF1ab
Amino acid change: Leucine → Phenylalanine (L1130F)
Samples with variant: 1,357
Mean CT with variant: 21.34
Mean CT without variant: 34.03
p-value: < 2.2e-16 (effectively 0)

This study investigated genetic variants associated with virulence in SARS-CoV-2, the causative agent of COVID-19. A total of 272,726 whole genome sequenced clinical isolates were analyzed, collected from patients in the UK by the COVID-19 Genomics UK (COG-UK) consortium between November 29th, 2021 and January 15th, 2022. They were each sequenced on the Illumina NovaSeq 6000 platform. Variants were called by aligning each genome against the Wuhan-Hu-1 reference strain (NC_045512) using MUMmer's dnadiff wrapper. This yielded 26,726 unique variants across all samples. Cycle threshold (CT) values were used as the virulence phenotype. CT values represent the number of PCR cycles required to detect viral RNA in a patient's nasal swab, where lower values indicate higher viral burden. A Student's t-test was performed for each variant comparing CT values among samples carrying the variant to those without it.

The variant most significantly associated with higher virulence was 3653_C_T, a C to T substitution at genome position 3653. This variant was identified in 1,357 samples and was located in ORF1ab, where it causes a missense mutation resulting in a Leucine to Phenylalanine substitution at amino acid position 1130 (L1130F). Samples carrying this variant had a mean CT of 21.34 (SD = 9.92), compared to a mean CT of 34.03 among samples without the variant. This corresponds with a difference of nearly 13 cycles, which is a roughly 8000 fold greater viral RNA burden. The t-test comparing CT values between the two groups yielded a p-value indistinguishable from 0, $p < 2.2\times10^{-16}$, indicating an extremely strong and statistically significant association between this variant and higher viral load.